

clients  $k \in S_t$ : local SGD  $\rightarrow \Delta_k^t$

fixed coordinate sketch  
 $\mathbf{z}_k = (\Delta_k)_I$

norms  $\|\Delta_k\|$ ,  
median  $M_t$

four statistics: alignment  $u_{k,1}$  · peer agreement  $u_{k,2}$   
magnitude regularity  $u_{k,3}$  · temporal stability  $u_{k,4}$

type-1 pass  $\rightarrow$   
heterogeneity  $\hat{s}^t$

FOU factor  $\varphi^t$   
(directional sets only)

interval type-2 inference (16 rules)  $\rightarrow$  exact  
centre-of-sets type reduction  $\rightarrow \hat{\tau}_k$

reputation memory  
 $\tau_k \leftarrow (1 - \rho)\tau_k + \rho\hat{\tau}_k$

$w_k \propto \pi_k e^{\tau_k/T}$   
(entropic allocation)

clip to  $\nu M_t$ , average,  $\theta^t = \theta^{t-1} + \sum_k w_k \tilde{\Delta}_k$

next round